Agent / Python Basics

1. Complete the notebook *as is* and:
    a. Ask the LLM a non-Chemistry question.
    b. Ask a chemistry-related question that involves solving a problem.
    c. Ask a question that involves remembering something earlier in the chat.

2. In the cell labelled 'set-up model:'
    a. Adjust the system prompt to:
        i. Give the LLM a personality.
        ii. Give the LLM rules for what it can and can't answer.
    b. In each of the cases in (2.a), see if the agent follows your additions (re-run that cell before re-running the chat loop.
3. In the cell labelled 'chat loop:'
    a. Change the prompts that appear when you and the LLM exchange messages to be something more interesting.
    b. The first non-comment line of code in this cell initializes the messages list. This is the list that gives the LLM memory, as it stores every conversation turn. In the while loop, add an if statement that resets the messages list to just the system prompt if the user types 'reset.'
    c. Test the addition made in (b) above.
    d. The tokenizer changes words to numbers for the LLM to read, and then changes the LLM's output numbers to words for you to read. After the comment #3, insert code that:
        i. Prints a line of = signs.
        ii. Prints the LLM output before the tokenizer changes it into words (this would be output_ids[0][inputs["input_ids"].shape[1] :]).
        iii. Prints another line of = signs.

        Now when you chat, you will see the LLM's actual output (a bunch of numbers) and the decoded words below. Test this.